MA'ADEN معادن

MA'ADEN ENGINEERING MANUAL

Geomembranes

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1. INTRODUCTION

1.1 Scope

This document specifies the minimum technical requirements for the supply, installation, and maintenance of geomembranes for all Ma'aden projects.

All materials shall meet or exceed the requirements of this specification, and all Work shall be performed in accordance with the procedures provided in this specifications.

This document is intended to be used in conjunction with the applicable documents listed herein.

1.2 Purpose

The purpose of this specification is to set the minimum engineering standard for supply, installation, and maintenance of geomembranes for all Ma'aden projects.

The selection of the type of geomembrane is critical to ensuring the desired life under the anticipated service conditions.

A Project specification must be prepared by the designing engineer in compliance with this Master Specification once they have determined the type, thickness and specific properties of the geomembrane to meet the field conditions.

2. KEY TERMS AND DEFINITIONS

Contractor	An entity, engaged or approved by the Owner, entrusted with a contract to perform Work or cause Work to be performed.
EPDM	Ethylene Propylene Diene Terpolymer
Employer	Saudi Arabian Mining Company (Ma'aden) and its authorized representatives and respective affiliates.
fPP	Flexible Polypropylene, Non-Reinforced
fPP-R	Flexible Polypropylene, Reinforced
GRI	Geosynthetic Research Institute
HDPE	High Density Polyethylene
LLDPE	Linear Low Density Polyethylene
OIT	Oxidative Induction Time
PVC	Polyvinyl Chloride
QA	Quality Assurance
QC	Quality Control
Site	Geographical location where Services are to be performed.
Work	The entire Work and Services and all things which the Contractor is or may be required to execute, supply and do under the Contract having regard to the Scope of Project, or to cause to be executed, supplied or done, for the purposes of constructing and Commissioning the Plant.

3. REFERENCED DOCUMENTS

This specification requires compliance with the provisions of the latest revision of the relevant standards and codes of the following authorities and technical organizations, as applicable, and specifications which shall be considered to form part of this specification, where applicable.

3.1 Statutory Regulations

All Work performed and materials furnished shall comply with the applicable statutory regulations, codes and other requirements of the Kingdom of Saudi Arabia (including but not limited to directives of the High Commission for Industrial Security, Royal Commission regulations applicable to the Site, regulations of the Presidency of Meteorology and Environment, Saudi Building Code and other applicable codes and regulations). The Contractor shall ensure that he is fully aware and informed of all Government laws, local ordinances, regulations and all orders and decrees of bodies or tribunals having any jurisdiction or authority over the Project and/or Site, which in any manner affect the Project and those engaged or employed on the Project, or which in any way affect the conduct of the Work. The Contractor shall at all times observe and comply with all such Government and/or Ministry laws, bylaws, ordinances, codes, regulations, orders and decrees.

3.2 Industry Codes and Standards

All equipments and systems shall conform to the following.

3.2.1 International Codes and Standards

3.2.1.1 American Society for Testing and Materials (ASTM)

ASTM D413	Standard Test Methods for Rubber Property – Adhesion to Flexible Substrate
ASTM D570	Standard Test Method for Water Absorption of Plastics
ASTM D638	Standard Test Method for Tensile Properties of Plastics
ASTM D696	Standard Test Method for Coefficient of Linear Thermal Expansion of Plastics between -30ºC and 30ºC with a Vitreous Silica Dilatometer
ASTM D751	Standard Test Methods for Coated Fabrics
ASTM D792 0150	Standard Test Methods for Density and Specific Gravity (Relative Density) of Plastics by Displacement
ASTM D882	Standard Test Method for Tensile Properties of Thin Plastic Sheeting
ASTM D883	Standard Terminology Relating to Plastics
ASTM D1004	Standard Test Method for Initial Tear Resistance (Graves Tear) of Plastic Film and Sheeting
ASTM D1149	Standard Test Method for Rubber Deterioration – Cracking in an Ozone Controlled Environment
ASTM D1203	Standard Test Method for Volatile Loss from Plastics Using Activated Carbon Methods
ASTM D1204	Standard Test Method for Linear Dimensional Changes of Non Rigid Thermoplastic Sheeting or Film at Elevated Temperature

ASTM D1238	Standard Test Method for Flow Rates of Thermoplastics by Extrusion Plastometer
ASTM D1505	Standard Test Method for Density of Plastics by Density-Gradient Technique
ASTM D1593	Specification for Nonrigid Vinyl Chloride Plastic Sheeting
ASTM D1603	Standard Test Method for Carbon Black in Olefin Plastics
ASTM D1693	Standard Test Method for Environmental Stress – Cracking of Ethylene Plastics
ASTM D1790	Standard test Method for Brittleness Temperature of Plastic Sheeting on Impact
ASTM D1822	Standard Test Method for Tensile-Impact Energy to Break Plastics and Electrical Insulating Materials
ASTM D3700	Standard Practice for Containing Hydrocarbon Fluid Samples Using a Floating Piston Cylinder
ASTM D3895	Standard Test Method for Oxidative Induction Time of Polyolefins by Thermal Analysis Differential Scanning Calorimetry
ASTM D4218	Test Method for Determination of Carbon Black Content in Polyethylene Compounds by the Muffle-Furnace Technique
ASTM D4437	Standard Practice for Determining the Integrity of Field Seams Used in Joining Flexible Polymeric Sheet Geomembranes
ASTM D4833	Test Method for Index Puncture Resistance of Geotextiles, Geomembranes and Related Products
ASTM D5199	Standard Test Method for Measuring the Nominal Thickness of Geosynthetics
ASTM D5397	Standard Test Method for Evaluation of Stress Crack Resistance of Polyolefins Geomembranes using Notched Constant Tensile Load Test
ASTM D5514	Standard Test Method for Large Scale Hydrostatic Puncture Testing of Geosynthetics
ASTM D5596	Standard Practice for Microscopical Evaluation of the Dispersion of Carbon Black in Polyolefin Geosynthetics
ASTM D5617	Standard Test Method for Multi-Axial Tension Test for Geosynthetics
ASTM D5641	Standard Practice for Geomembrane Seam Evaluation by Vacuum Chamber
ASTM D5721	Practice for Air-Oven Aging of Polyolefin Geomembranes
ASTM D5820	Standard Practice for Pressurized Air Channel Evaluation of Dual Seamed Geomembranes

ASTM D5885	Test Method for Oxidative Induction Time of Polyolefin Geosynthetics by High Pressure Differential Scanning Calorimetry
ASTM D5994	Standard Test Method for Measuring Core Thickness of Textured Geomembranes
ASTM D6365	Standard Practice for the Non destructive Testing of Geomembrane Seams using the Spark Test
ASTM D7176 – 06	Standard Specification for Non-Reinforced Polyvinyl Chloride (PVC) Geomembranes Used in Buried Applications
ASTM D7177 – 05	Standard Specification for Air Channel Evaluation of Polyvinyl Chloride (PVC) Dual Track Seamed Geomembranes
ASTM D7408 – 08	Standard Specification for Non Reinforced PVC (Polyvinyl Chloride) Geomembrane Seams

3.2.1.2 Geosynthetic Research Institute Standards (GRI)

GRI GM 10	Specification for the Stress Crack Resistance of Geomembrane Sheet
GRI GM 13	Test Properties, Testing Frequency and Recommended Warranty for High Density Polyethylene (HDPE) Smooth and Textured Geomembranes
GRI GM17	Test Properties, Testing Frequency and Recommended Warranty for Linear Low Density Polyethylene (LLDPE) Smooth and Textured Geomembranes
GRI GM18	Test Methods, Test Properties and Testing Frequencies for Flexible Polypropylene, Nonreinforced (fPP) and Reinforced (fPP-R) Geomembranes
GRI GM21	Test Methods, Properties, and Frequencies for Ethylene Propylene Diene Terpolymer (EPDM) Nonreinforced and Scrim Reinforced Geomembranes

3.2.2 Alternative Codes

Alternative codes may be used for supply, installation, and maintenance of geomembranes if approved by the Employer.

3.3 Ma'aden Standards

MD-101-SMPM-PM-QA-PEI-0001 Contractor Quality Requirements

3.4 Site Data

All materials, equipment and work methods shall be suitable for the Site conditions as defined in the applicable Site Data Specification.

4. MATERIALS

4.1 Materials Specification

The Contractor shall demonstrate that the materials meet the standards set forth by this Specification by testing of the lot samples as set out herein and by further actions as necessary should there be any grounds for doubt.

4.2 Geomembrane Materials Compatibility with the Impounded Material

The chemical properties of the impounded material shall be detailed similar to Table A-1 in Appendix A, a compatible geomembrane lining system shall be specified.

As per Section 7.3, the Manufacturer shall provide a written warranty that the geomembrane, when handled and installed in accordance with the Manufacturer's recommended procedure, shall perform without deterioration when subject to the in-service conditions for the "Required Service Life" (refer Table A-1 in Appendix A).

The Manufacturer shall also provide an assessment of the anticipated change in material properties beyond the "Required Service Life" (refer Table A-1 in Appendix A).

For the purposes of this section of the specification, deterioration is defined as any significant increase in permeability of the geomembrane material or any change in the chemical and/or mechanical properties exceeding 25% of the specified values.

4.3 Material Requirements

Except where specifically allowed in the specifications referenced in Section 3.2.1, all geomembranes supplied shall be manufactured from new, first quality products. Where permitted by the abovementioned specifications, recycled polymer shall be from the same manufacturing process and within the limits specified therein. Geomembranes supplied shall be free of holes, blisters, thin spots, un-dispersed raw materials, or any sign of contamination by foreign matter, designed and manufactured specifically for sustaining the in-service conditions specified in the relevant contract specifications. The type, texturing and thickness of geomembrane to be used shall be as nominated in the relevant drawings and/or contract specifications.

Addition of all ingredients in the mixing stage shall be such as to ensure complete dispersion throughout the compound prior to manufacture. It shall be uniform in color and surface texture and consistent in thickness.

The specified geomembrane material shall meet or exceed the values stated in the applicable Geosynthetic Research Institute or ASTM standards.

The geomembrane material shall be suitable for thermal fusion and hot air seaming methods.

4.4 Materials Quality and Testing

The geomembrane material components and manufacture, quality control and testing shall comply with the Geosynthetic Research Institute or ASTM standards and the test methods outlined in the applicable Standards.

Section 7.2 specifies the need for documentation to be submitted prior to delivery of material to Site to avoid potential delays.

The Contractor shall submit the QA test certificates for each roll of geomembrane supplied to the project. The test results shall be reported relative to the roll numbers and at the minimum frequencies specified in the applicable Geosynthetic Research Institute or ASTM standard.

The Employer may decide to conduct third party testing on samples recovered from rolls supplied to Site. The results of third party results shall take precedence over the test results from the Contractor.

4.5 Latent Defects

Fisheyes are referred to as the thinning of the geomembrane. The presence of Fisheyes in geomembrane is of concern regardless of its size. Fisheyes often occur repetitively down the length of the roll and at the same location from the roll edge. Its presence is clearly related to a manufacturing problem.

The maximum acceptable manufacturing defect is one per 300 m^2 of roll, which shall be patched as part of the installation of the geomembrane. Where more than the maximum allowable defects occur, the Contractor shall be liable to replace the geomembrane roll or part roll.

Surface blemishes are considered to be areas where the resin has not been properly mixed or filler can be noted on the geomembrane. The color and surface texture of the geomembrane shall be uniform and consistent. If the area is rejected as unacceptable, then the section of the geomembrane roll shall be removed and replaced, and be treated as a manufacturing defect.

Straight-line scratches that penetrate the geomembrane (approximately 0.1 mm) is a defect. The roll length where such a defect occurs shall be cut out and removed from Site. The Contractor shall be liable for the replacement of the defective section of geomembrane.

Where it becomes apparent that a batch of geomembrane rolls related to manufacturing machine number, roll sequence or resin type, are defective the whole batch shall be rejected. The Contractor shall be liable to replace the rejected batch at his own expense.

4.6 Geomembrane Delivery and Acceptance

The geomembrane material shall be led and fixed onto suitable forms or cores and wound to give a uniform tension across each roll such that:

- End impressions shall not occur
- During lifting the roll does not deform to form creases in the geomembrane
- Any relaxation or material shrinking after unrolling shall be less than 0.5%
- On unwinding and following any relaxation or shrinkage, the final useable length of the material roll shall not be less than the specified roll length, and
- On unwinding and following any relaxation or shrinkage, the material shall lay naturally flat and straight and be stable and free from edge cockling, banana shaping and creasing.

Each roll shall be labeled so as to provide the following identifying data:

- Product Code
- Product type and Description
- Roll Number
- Batch Number/Order Number
- Width

- Thickness, and
- Roll Length.

This data shall be provided prior to, or with, delivery of all materials to Site. All materials shall be certified by the manufacturer's laboratory test reports prior to installation.

5. MANUFACTURER SPECIFICATION

5.1 Manufacturer's Qualifications

The manufacturer of geomembranes of the type specified shall have at least five years experience in the manufacture of such geomembranes. In addition, the geomembrane manufacturer shall have manufactured at least 1,000,000 m^2 of the specified product in the last five years.

5.2 Submittals

The Contractor shall submit the following to the Employer at the time of placing the order for the Geomembrane with the Geomembrane supplier.

5.2.1 Polyethylene Resin Submittals

- Certification that any polyethylene resin is new first quality resin manufactured from virgin uncontaminated ingredients which meet or exceed the requirements set out herein
- Certification that all resin is from the same manufacturer
- Test certificates on resin to meet GRI minimum standard for Oxidative Induction Time (OIT). (Minimum 100 minutes).

5.2.2 Manufacturer Submittals

- Certification of production capacity and schedule availability to meet the requirements of the projects
- Origin identification and productions date(s) of the raw materials used to manufacture the geomembrane for the project
- Copies of the quality control certificates of the raw materials used to manufacture the geomembrane for the projects
- Certified test results for all material properties as specified
- Certification that no external reclaimed polymer and no external off work material is added to the virgin resin during the manufacture of this geomembrane. Addition of recycled polymer from the same manufacturing process shall not exceed the limits allowed in the applicable GRI specification, or if not specified by GRI, shall not exceed 2% by weight.
- A listing of at least five separate facilities using this geomembrane totalling a minimum of 2,000,000 m^2 and including at least 1,000,000 m^2 of this same grade of geomembrane. Provide full details including project, location, name of contact person, phone numbers, address, size of project, etc.
- Manufacturer to submit the type of stabilizers and anti oxidants to be used in the manufacture of the geomembrane to be supplied. The Employer will treat this information as confidential.

6. EXECUTION

6.1 Installation Layout

The Contractor shall submit a detailed layout clearly defining the sheet layout, all field welds, corner arrangements, weld crossovers, offsets and base-to-embankment weld locations for review by the Employer prior to commencement of installation works. Any changes, deviations, tests and repairs shall only be made following written approval of the Employer and shall be fully detailed on the "As-Built drawings".

The geomembrane and weld layout shall take into account the following requirements:

- Materials shall be installed in continuous sheets down embankments or across the base and shall overlap adjacent geomembrane sheets by a minimum of 100 mm or to any additional width required by the type of welding equipment employed
- Embankment lining shall extend a minimum of 2.0 m from the toe into the base before any perimeter base welds are permitted
- No horizontal field welds shall be permitted within the embankment area and due care shall be given to ensuring that welds are orientated directly up the embankment slopes. In the corners it may be necessary to have short field seams running close to horizontal, however such seams shall be approved by the Employer prior to installation of the geomembrane.
- The geomembrane shall terminate on the bottom of the anchor trench located at the top of the embankment as detailed on the Drawings, and
- Cross-over or T joint welds must be offset a minimum of 300 mm to avoid multi-layer weld lapping.

The Contractor shall be responsible for making any allowances considered necessary to accommodate predictable differential settlements of the sub-grade, and/or variations in temperature. In particular, allowances shall be made to accommodate the ultimate thermal movement of the geomembrane during the coldest periods in the region.

6.2 Factory Fabrication (Where Applicable)

6.2.1 General

Factory fabrication shall be carried out wherever possible, for example in PVC geomembrane, which can be pre-fabricated into large panels, folded and deployed on Site. Factory fabrication shall be performed by experienced personnel under constant supervision by the Contractor.

The Contractor shall ensure that at all times the fabrication facilities are kept clean and dust free.

All factory seams shall be inspected and tested over the full seam length and width prior to packaging by the Contractor. The Employer reserves the right to accompany the Contractor and carry out joint testing of the seams. Each seam shall be initialed by the Contractor with indelible ink. Close inspection shall be made to ensure there are no loose edges within factory seams. Factory seams shall not cause excessive curvature over the length of the panel.

The Employer will inspect, at his discretion, the fabrication facilities, procedures and quality of workmanship within the factory at any and all times during the fabrication process of geomembrane panels to ensure the required standards as detailed in the Specification are performed.

During the fabrication of the geomembrane panels the Employer will have complete authority to order immediate stoppage of the Work or rejection of seams and material damaged due to the use of improper fabrication procedures or any reason which, in his sole opinion, may result in a defective geomembrane.

The factory seams shall meet or exceed the specification values shown in Table A-1 in Appendix A.

6.2.2 Panel Size and Layout

Individual widths of calendared material shall be fabricated into panels of maximum practical size prior to delivery on Site.

Panels shall be fabricated such that all edge joints run vertically or near vertically up slopes. End joints shall be staggered and kept to a minimum.

The Contractor shall fabricate panels to the size and number as indicated on the installation contract layout drawing which shall be agreed to by the Contractor and the Employer prior to Contract award.

6.2.3 Factory Seaming

Factory seaming shall be carried out using hot air welding.

Contact surfaces of material to be seamed shall be wiped with specified rags until all foreign matter, dust, dirt, etc have been removed.

The equipment used for hot air welding of seams shall be self propelled (but aligned manually) and have suitable controls for adjustment of the operating temperature and speed of travel along the seam. This equipment shall also be of the type which raises the geomembrane off the ground, in the area of the seam and applies pressure to the seam by upper and lower rollers.

To ensure the correct weld is achieved within seams, prior to use each item of equipment used for hot air welding shall be calibrated and set for operating temperature, speed of travel along the seam and pressure applied to the seam by the rollers.

Alternative seaming methods to hot air welding will be considered provided sufficient information is supplied with the tender to enable the proposed method to be adequately assessed.

All necessary precautions shall be taken to prevent the occurrence of 'blocking', which is unintentional adhesion that may occur during storage or shipping between plastic films or between a film and another surface. (Refer ASTM D883).

Factory seaming shall also comply with the requirements of Section 6.4.4.

6.3 Sub-Grade Preparation

All sub-grade surfaces over which the geomembrane are to be placed shall be prepared to the standards specified herein.

- The surface (termed "liner sub-grade") upon which the geomembrane is to be placed shall be constructed from well-graded material, free of organics, trash, clay-balls, or other deleterious material that may cause damage to the geomembrane. Unless otherwise indicated in the Drawings or by the geomembrane supplier, the upper 150 mm of the finished liner sub-grade shall not contain stones or debris larger than 10 mm. The liner sub-grade shall be compacted in accordance with design specifications. But, in no event shall the compaction be less than the level that is required to provide a firm unyielding foundation sufficient to permit the movement of vehicles and welding equipment over the liner sub-grade without causing rutting or other deleterious effects. The liner sub-grade shall have no sudden sharp or abrupt changes in grade.
- The surface upon which the geomembrane is to be placed shall consist of a flat, smooth, surface free of foreign matter, sharp, irregular or abrupt changes in line and level
- The Contractor shall inspect the surface of each area with the geomembrane installation sub-contractor and the Employer to confirm the surface treatment required (if any) before any geomembrane is installed. If required, additional treatment of the sub-grade shall be carried out by the Contractor at the direction of the Employer.
- The Contractor shall ensure that any prepared area is protected from stormwater run-off damage in the event of rain prior to installation of geomembrane
- The Contractor shall ensure that in the event of rain stormwater is not able to penetrate between the installed geomembrane and the sub-grade.

6.4 Field Installation

6.4.1 Geomembrane Transport and Handling

Geomembrane materials shall be stored at the Site in an area designated by the Employer.

The Contractor shall be responsible for all freight, transportation to Site and handling and storage on Site, including Site security unless otherwise specified.

Materials shall be stored in such a manner that no surface irregularities or other influences will apply point loads, abrasions, cuts or distortions or other forms of damage to the rolls.

No more than two rolls of geomembrane shall be stacked one on top of the other. The Employer may direct the Contractor to make adjustments to the stored materials if evidence of damage is observed. Any defects or damage detected shall be repaired to acceptable standards or totally replaced by and at the Contractor's expense.

6.4.2 Geomembrane Installation

The geomembrane shall be installed as shown on the drawings and in accordance with the approved layout drawing.

The Contractor shall inspect the entire surface area of each and every sheet during unrolling and placement to ensure that there are no tears, abrasions, indentations, cracks, thin spots or other faults in the material. The materials shall meet the specifications and requirements as outlined in Section 5. Sheets shall be rejected where manufacturing process irregularities are evident including irregular thickness and imperfections that will not provide even stress distribution throughout the sheet. The Contractor shall inform the Employer of any such occurrence. It shall be the responsibility of the Contractor to ensure that any such damage or imperfections are repaired or replaced in accordance with the requirements of this Specification.

The geomembrane shall be placed in a relaxed state over the prepared surface allowing the material to respond to thermal changes without causing excessive buckling, wrinkling or tensioning. No "fishmouths" or other signs of stress shall be permitted within welds made during installation of the geomembrane. Any such occurrences being identified shall be repaired or remedied by the Contractor at no cost to the Employer. The Contractor shall be responsible for making any allowances considered necessary to accommodate predictable differential settlements of the surface and/or variations in temperature.

The geomembrane shall be anchored along the embankment crest within backfilled and compacted anchor trenches, as shown on the drawings. Material for the backfilling of the anchor trenches, shall be cohesive material and subject to the approval of the Employer. The backfill material shall be placed and compacted to 95% of Modified Proctor Dry Density at ±2% Optimum Moisture Content unless otherwise noted in the contract specification.

6.4.3 Weather Conditions for Field Seams

Many factors, such as the geomembrane temperature, humidity, wind, precipitation can affect the integrity of field seams and must be taken into account when deciding whether or not seaming should proceed.

Test seams shall be required prior to daily production seaming to determine if the weather conditions will affect the Contractor's ability to produce quality seams. If the test seams fail to comply with the specifications and if in the opinion of the Employer such failure is due to prevailing weather conditions, field seaming shall not be commenced until weather conditions have improved and the Contractor can demonstrate to the Employer that the quality of seams will not be adversely affected by the weather. The Contractor shall be responsible for all expenses associated with delays due to prevailing weather conditions.

6.4.4 Field Welding

6.4.4.1 General Requirements

The geomembrane sheets shall be of such lengths and widths and shall be placed in such a manner so as to minimize field seaming.

Horizontal field welds on slopes shall not be permitted and the geomembrane shall lay absolutely flat when welded. All sheets shall be installed in a relaxed state and shall be free of tension or stress upon completion of installation. Stretching of the geomembrane to fit shall not be permitted.

Welding shall commence as soon as practical after the sheet is deployed.

No personnel working on the geomembrane shall wear spiked shoes or shoes with sharp protrusions etc. or other materials/implements that can damage the geomembrane or engage in actions that could result in damage to the geomembrane.

The geomembrane shall be deployed in a manner to minimize wrinkles.

Any area of a sheet seriously damaged (torn, twisted, or crimped) shall be marked and repaired in accordance with this Specification.

Field joints shall be fabricated using hot air or hot wedge welding equipment and installation methods as described in this Specification and to the manufacturer's recommendations. Details of all materials and equipment to be used shall be submitted to the Employer for approval prior to commencement of installation.

Joints shall be made so as to form a longitudinal sealed air channel to allow visual checking, or air pressure testing prior to acceptance.

All geomembrane placed on slopes shall be extended onto the base area without transverse seams.

6.4.4.2 Hot Air Welding

The equipment used for hot air welding of seams shall be self-propelled (but aligned manually) and have suitable controls for adjustment of the operating temperature and speed of travel along the seam. This equipment shall also be of the type that raises the geomembrane off the floor in the area of the seam and applies pressure to the seam by upper and lower rollers.

To ensure the correct weld is achieved within seams, each item of equipment used for hot air welding shall be calibrated and set prior to use, for operating temperature, speed of travel along the seam and pressure applied to the seam by the rollers to make it satisfactory and compatible for the successful welding of the material.

The welding machine shall incorporate rollers that produce a 10 mm wide channel for air pressure testing of seams.

6.4.4.3 Seam Construction Joints

Care must be taken in applying hot air to each seam construction joint at the ends of an increment of seam previously completed, so that leak paths or weak points in the seam do not occur at these joints.

6.4.4.4 Preparation

Contact surfaces of material to be seamed shall be wiped with clean, dry, white, absorbent, lint free cotton rags until all foreign matter, dust etc. have been removed. When a rag shows discoloration from use, it shall be discarded and replaced with a fresh one.

When conditions are harsh, such as in high heat or high winds, or when other conditions increase the possibility of dirt or foreign material being deposited on the contact surfaces, the time between wiping and seaming shall be varied as necessary to produce acceptable seams.

Individual sheets shall be laid out and overlapped to a minimum of 100 mm prior to welding.

6.4.4.5 Fishmouths

'Fishmouths' are wrinkles in the geomembrane, which occur in seams.

Extreme care shall be taken through the Work to avoid fishmouths in the field seams. Fishmouths can usually be eliminated by gently tugging on the seam just completed in the direction of the completed seam. Unless the Contractor gives constant attention to this item, fishmouths will be a frequent occurrence.

Where fishmouths do occur, they shall be slit far enough from the seam to dissipate them, lapped, seamed together in the lapped area, and patched in accordance with this Specification.

6.4.4.6 Butt Joints

Lap joints shall be used wherever possible. Should butt joints be unavoidable they shall be formed using a joint cover strip of the same material as the geomembrane and of 100 mm minimum width for extricable seams and 50 mm for hot air seams. Materials joined shall be butted together and the joint cover strip shall centrally straddle the joint and be fully bonded across its entire width in accordance with the above procedure for lap joints. No free edges shall be permitted on the joint cover strips.

6.4.4.7 Transverse Joints

When a transverse joint is encountered within a seam, the free edges on the bottom of the overlapping sheet shall be sealed for a minimum of 150 mm from end of the sheet prior to making a transverse seam.

All intersections of joints shall be inspected and tested using a Vacuum Box in accordance with this Specification.

6.4.4.8 Patching

All punctures, cuts, tears, severe abrasions and similar damage or abuse suffered by the geomembrane material shall be repaired by patching and the location, time and extent of the patching noted by the Contractor.

Areas requiring repair or patching shall be prominently identified by suitable marking applied to the sheet material outside the area to be repaired or patches so as not to introduce marking substances into the seam area.

Patches shall be cut from flat, unwrinkled scraps of geomembrane material and shall be free of defects, field seams and factory seams. They shall be of sufficient size to extend a minimum of 50 mm in all directions beyond the limits of any puncture, cut, tear, abrasion etc.

Patches shall be applied as specified above for welded joints. The parent material shall be gently pulled and held flat in the area to be patched so as to provide an acceptable surface to receive the patch. The applied extrusion weld shall be continuous around the patch.

Where patches over fishmouths are required, they shall extend a minimum of 100 mm each side of the edge of the overlap, and a minimum of 100 mm beyond the ends of the cut in the fishmouth.

All patches and repairs shall be inspected and tested using a Vacuum Box as detailed in this Specification.

6.4.4.9 Holding Down of Geomembrane

Sandbags shall be used to hold down the geomembrane during installation. Sandbag material shall be durable and waterproof, shall not have any adverse effects on the geomembrane, and will be subject to the approval of the Employer. The Contractor shall install sufficient sandbags to prevent uplift or movement of the geomembrane during the entire course of the Works.

The Contractor shall implement measures to ensure the safety of Site personnel in the advent that windy conditions arise during the Works. Such measures may include venting of the geomembrane as well as provision of additional sand bags or other weights, particularly on leading edges. Such measures shall be planned, documented and implemented subject to the approval of the Employer. Ad-hoc measures shall not be acceptable and may result in the Employer standing down installation work until satisfied that appropriate measures are in place to ensure personnel safety. Expense resulting from such stand down shall be to the Contractor's account.

6.4.4.10 Seals to Concrete Sumps and Pipe Penetrations

Where specified, the geomembrane shall be joined and sealed to the pipe penetration to form a leak proof seal.

All contact surfaces shall be free of all dirt, dust, free moisture, oil and other foreign materials prior to the joint being made.

6.5 Care and Protection of the Works

During installation of the geomembrane, every precaution shall be taken to prevent damage to the Works. It shall be the responsibility of the Contractor to report any damage or acts which may have caused damage to the geomembrane.

Properly protected tools and equipment shall be used at all times.

Trafficking of personnel or equipment over the geomembrane shall be kept to the minimum necessary to complete the Works and then only in areas where adequate precautions have been taken.

Tarpaulins of geomembrane material of suitable size shall be spread out on the geomembrane as work areas for preparing patches, storing tools, supplies, rags, etc. Under no circumstances shall petrol-driven generators or cans of solvent or petrol be placed directly on geomembrane.

At no time are cutting knives or similar to be used in the installation of the geomembrane unless they have a hook blade and have been approved by the Employer.

6.6 Clean-Up

Clean-up within the lined area shall be an ongoing responsibility of the Contractor throughout the course of the Works. Particular care shall be taken to ensure that no rubbish, tools and other unwanted materials are trapped beneath the geomembrane. Care shall be taken to ensure that all scraps of geomembrane material are removed from the Works on completion of installation of the geomembrane.

7. INSPECTION AND TESTING

7.1 Quality Assurance Plan

The Contractor shall provide a project specific Quality Assurance/Quality Control Plan (QA/QC Plan) prior to the commencement of the Work that includes the following elements:

- A manufacturer's QA/QC program that includes raw materials control, storage and handling, process control and finished product control in accordance with the requirements of ISO 9001
- A field installation QA/QC program that includes deployed membrane panel inspection, Site and surface preparations, alignment, welding process control, repair, and non-destructive and destructive testing in accordance with the requirements of ISO 9001.

The Contractor shall nominate a person in the QA/QC Plan who is to act as Site Quality Control Officer and who shall coordinate all testing and shall retain all results, reports and certificates as they are generated. The person nominated shall be subject to prior approval of the Employer and shall have demonstrated experience in installation of more than 500,000 m^2 of geomembrane, or such other level of experience deemed sufficient by the Employer.

As a minimum, the QA/QC Plan shall contain the following items:

- Sheet Layout Drawing clearly defining the sheet layout, all field welds, corner arrangements, weld crossovers, offsets and base-to-embankment weld locations
- Details of the materials and equipment to be used for the field welding
- Details of the experience and qualifications of the nominated Site Quality Control Officer

- Details of the testing procedures and plan as per this Specification and any variations the Contractor proposes, and
- All other items relevant to the supply, installation and testing of the geomembrane material in accordance with this Specification.

7.2 Manufacturer's Quality Assurance Report

It is a requirement that all geomembranes be certified for quality prior to installation. For each roll, the Contractor shall submit a separate Manufacturer's Quality Assurance Report to the Employer. This report shall comprise a Raw Materials Batch Report and a Product Report.

The Batch Report shall be provided for each and every raw material batch associated with the specific geomembrane product delivered to Site and shall conform to the following:

- Density Test ASTM D1505
- Melt Index ASTM D1238

The Product Report shall contain data compiled for each and every roll of geomembrane delivered to Site and shall include the information specified in the relevant GRI or ASTM specification for the supply of the geomembrane.

The Manufacturer's Report (along with the Materials Identification) shall be submitted prior to the arrival of the material to the Site. No materials will be accepted for delivery to Site, or progress payment made, unless all necessary Manufacturer's Quality Assurance certification data, as described above, has been provided to the Employer.

All such data must be supplied in sufficient time such that no delay shall be caused to the project program. Failure to conform to this requirement, causing resultant delay to the progress of the Works, will not be grounds for an extension in time or removal of any commercial penalties that accompany this Contract.

7.3 Warranty of Performance

The Contractor shall provide a written warranty stating the duration of time that the geomembrane, when handled and installed in accordance with the Manufacturer's recommended procedure, shall perform without deterioration when subject to the in-service conditions. This duration of time, from the time of completion of installation, shall not be less than the time stated in Table A-1 in Appendix A. The Contractor shall also provide an assessment of anticipated change in material properties during the period of time quoted in the warranty.

The Contractor shall warrant the installation of the geomembrane for a period of one year from the date of completion of geomembrane installation and submit this in writing to the Employer.

7.4 Field Welds

7.4.1 Trial Welds

Trial welds shall be made on off-cut or fragment pieces of geomembrane in order to verify that adequate seams can be produced by the chosen equipment and settings in the prevailing conditions. It is important that the conditions under which these seams are made reflect the actual conditions influencing the seam welding.

Trial welds shall be made at the following times and carried out by the Contractor in the presence of the Employer:

- Prior to commencement of each shift
- Prior to each and every primary weld function
- Each alteration to extradite or thermal controls, and
- Following every 800 m of completed weld in the field.

Trial welds shall be at least 1.5 m in length and shall be clearly marked to identify the time of test, or prevailing environmental conditions that will influence performance of the weld. Each test weld shall be suitably labeled and identified so that all weld functions associated with the test area can be verified. In the event that a test is carried out in the course of executing a field weld, the type of weld and its location within the site works shall be clearly identified on the test strip.

7.4.2 Trial Weld Tensile Test

A tensile test shall be carried out in accordance with ASTM D 638.

The criterion for acceptance is the failure of the specimen by ductile failure in the sheet material prior to sheet weld failure. The test specimen shall achieve a minimum tensile value of yield of 80% of the parent sheet value.

7.4.3 Trial Weld Peel Test

The peel test shall be carried out in accordance with ASTM D 638.

The criterion for acceptance is the failure of the specimen by ductile failure in the sheet material prior to the sheet weld failure. The test specimen shall achieve a minimum tensile value of yield of 70% of the parent material value.

7.4.4 Non-Destructive Pressure Testing

At the completion of each and every primary weld the Contractor shall pressure test the entire length of these welds. All tests shall be carried out in the presence of the Employer unless otherwise approved.

The pressure test kit will consist of a manually operated air pump fitted with a pressure gauge of sufficient size and appropriate divisions to record the test pressures. Each end of the weld shall be sealed to prevent any air leakage from the weld zone for the duration of the test.

Where the weld test length exceeds 30 m, a second pressure gauge with valve shall be employed within the weld zone at the extreme opposite end of the test weld. The weld section shall be pressurized to between 170 kPa and 210 kPa. When the pressure has stabilized, the time should be noted. After a period of three minutes has elapsed, the pressures shall be recorded. At the completion of the test the pressure gauges will be removed from the weld zone and injection preparations suitably prepared and applied.

Where air loss in excess of 30 kPa over the three minutes is observed, the section shall be deemed to have failed. The faulty section shall be identified by successively testing shorter lengths. If the fault is of short length and isolated, it may either be repaired by the application of a patch or, where acceptable to the Employer, repaired by manual hot air or extradite fusion. Where more than one fault is found, the weld shall be reconstructed. The entire weld shall then be pressure tested again. If the second weld test passes, the weld will be accepted. In the event the second test fails, the Employer will direct that independent destructive testing of that entire weld be undertaken. The cost of this testwork shall be borne by the Contractor.

Acceptance of the weld following all tests in no way relieves the Contractor of his responsibilities regarding the quality of the Work and the workmanship employed.

The details of pressure tested primary welds shall be recorded on As-Built drawings and submitted to the Employer as required.

7.4.5 Vacuum Box Testing

Each and every secondary weld (i.e. overlap welds) shall be subjected to 100% continuous vacuum box testing and point stress testing as directed by the Employer. All tests shall be carried out in the presence of the Employer unless otherwise approved.

The vacuum box test shall apply to all welds, patches, test sites, and any other repair where extradite surface welding has been employed. No departure from this test will be allowed. The Contractor may wish to apply any additional non-destructive test method to the welds, however, the vacuum box test method will be mandatory.

Vacuum box testing shall also be carried out on each and every cross-over weld (primary or secondary) after pressure testing has been completed.

After obtaining a minimum of –20 kPa in the box for a period of 10 seconds, the seam should be inspected for the presence of bubbles. If bubbles are detected, the seam is deemed to have failed.

Where a fault within any weld is found, the weld shall be repaired by the Contractor (at no cost to Ma'aden) using a secondary weld method and then shall be re-tested using the vacuum box to ensure suitability. In the event the second test fails, the Employer will direct that independent destructive testing of that weld be undertaken. The repaired fault will be clearly detailed on the As-Built drawings.

7.4.6 Destructive Testing

The Employer shall select one test specimen for every 800 m of field weld performed for destructive testing. Each specimen shall be 1.5 m long (measured with the direction of the weld) x 0.3 m wide. The specimen shall be cut into two specimens of equal dimensions (cut across the weld direction) and the following testwork shall be carried out:

Specimen 1

Test Procedure: ASTM D 638

Type of test: 3 No. "Peel" tests
3 No. "Tensile" tests

An independent testing authority, as approved by the Employer, shall test Specimen 1. The Contractor shall be responsible for organizing this testwork, and the costs associated with the testing shall be borne by the Contractor.

A test certificate, prepared by the testing authority, shall be submitted to the Employer.

Specimen 2

Specimen 2 shall be tested by the Contractor on Site prior to any additional welding being carried out and as soon after the weld is made as possible. The testing will be carried out in the presence of the Employer.

The testwork on Specimen 2 shall comprise:

Test Procedure: ASTM D 638

Type of Test: 2 No. "Peel" tests

2 No. "Tensile" tests

A field test report, detailing the results of this testwork, shall be submitted to the Employer.

All test data shall be matched and qualified prior to that weld or geomembrane area being released for commissioning. Certification for release by the Employer shall in no way relieve the Contractor of its responsibility for the quality of the installation.

Where any two tests record failure, a further sample shall be extracted from an area adjacent to the specimen weld area and the same field and independent tests shall be carried out. Where these additional tests are positive, localized repairs (using secondary weld methods) shall be carried out as directed by the Employer. Where additional failure of the specimen results, the entire weld shall be removed and replaced. The repair of welds or replacement and subsequent testing, as required, will be carried out at no cost to the Employer.

The locations of all repair work shall be recorded on the "As-Built" drawings.

8. SAFETY

The Contractor shall take all necessary action to ensure the safety of personnel throughout the supply, shop fabrication, site fabrication and installation of the geomembrane. In particular attention shall be given to the safety of personnel and the integrity of the geomembrane system during windy conditions on Site.

9. AS-BUILT DRAWINGS

At the completion of the Works the Contractor shall submit to the Employer as-built panel layouts, trial welding logs, non-destructive test logs and destructive test logs as specified in the contract documents.

10. QUALITY CONTROL

The Contractor shall refer to the following.

MD-101-SMPM-PM-QA-PEI-0001 Contractor Quality Requirements

Appendix A

Service Conditions

Table A-1 : Service Conditions

Required Service Life	To be defined for each project
Physical Conditions	**Average/Maximum Value**
pH	To be defined for each project
Temperature at the geomembrane	To be defined for each project
Chemical Compounds	**Average/Maximum Value**
To be defined for each project	To be defined for each project
To be defined for each project	To be defined for each project
To be defined for each project	To be defined for each project
To be defined for each project	To be defined for each project
To be defined for each project	To be defined for each project
To be defined for each project	To be defined for each project